

Windows_Python_Serial_Messge_UIM2513_AN251219_Configuration and Compilation

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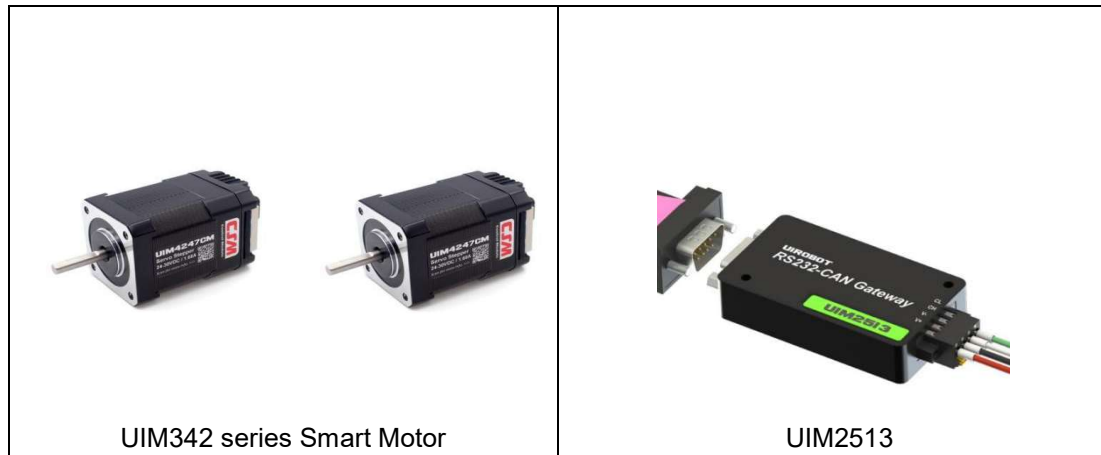
Background

Windows_Python_Serial_Messge_UIM2513 is a demonstration program based on UIROBOT UIM342 motor controllers to control stepper motors. This project demonstrates how to control multiple motors through CAN bus networks to achieve various motion control functions.

- Supports serial communication methods
- Multi-axis synchronous motion control
- Real-time status monitoring and feedback
- Sensor-triggered homing functionality
- Complete error handling mechanism

The code also supports Linux and macOS systems, refer to the README.md for further information.

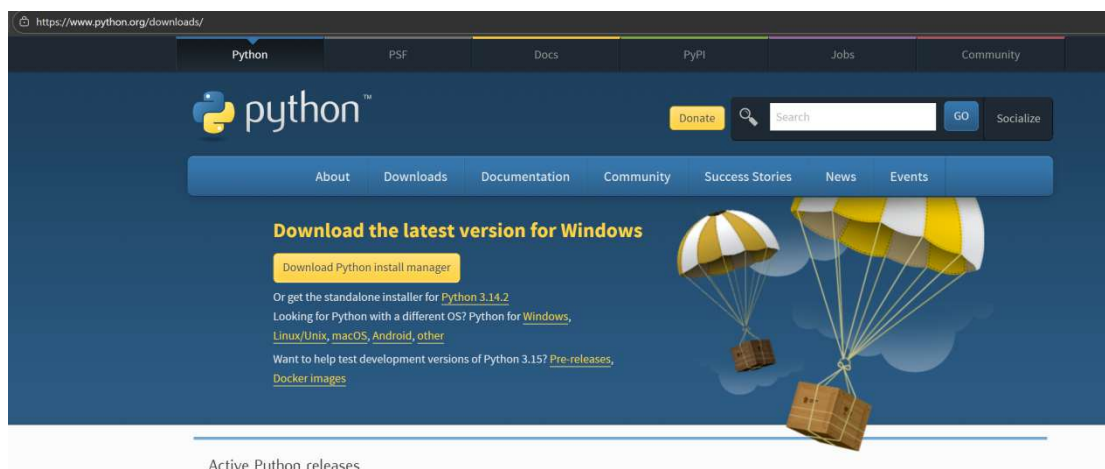
Required Materials



Downloading Python

To download Python on Windows, follow these steps:

- Visit the official Python website: <https://www.python.org/downloads/>
- On the download page, select a Python version suitable for Windows (Python 3.11 or higher is recommended).
- Click the "Download Python" button to download the Windows installer (.exe file).
- Note: When downloading, select the version that matches your system architecture (32-bit or 64-bit). Most modern Windows systems are 64-bit.



Installing Python

After downloading, follow these steps to install Python:

- Double-click the downloaded Python installer (.exe file) to launch the installation wizard.
- On the installation screen,
- **Important: Check the "Add Python to PATH" option so you can use Python commands directly from the command line.**
- Select "Install Now" for default installation, or choose "Customize installation" for custom installation.
- Wait for the installation to complete. The installation process may take a few minutes.
- After installation is complete, click "Close" to close the installer.
- Verify installation:
 - Open Command Prompt (CMD) or PowerShell and enter the following command:
`>> python --version`
if the Python version number is displayed, the installation was successful.

Installing Python Libraries

Python uses pip (Python package manager) to install third-party libraries. Here are the steps to install libraries:

- Open Command Prompt (CMD) or PowerShell.
`>> pip install pyserial>=3.5 pyserial-asyncio>=0.6`
- Installing multiple libraries from requirements.txt file:
`>> pip install -r requirements.txt`
- Upgrade pip (optional but recommended):
`>> python -m pip install --upgrade pip`

Running Python Program

After installing Python and required libraries, you can run Python programs in the following ways:

- Method 1: Using Command Line
 - Open Command Prompt (CMD) or PowerShell
 - Use the cd command to navigate to the directory containing your Python program:
`>> cd D:\Python_Serial_Messge_UIM2513_AN251219`
 - Run the main.py:
`>> python src\main.py`

The operating interface is shown in the figure.

```
Windows PowerShell
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安装最新的 PowerShell, 了解新功能和改进! https://aka.ms/PSWindows

PS D:\Windows_Python_Serial_Messge_UIM2513_AN251219> python.exe .\src\main.py
Auto-connecting to saved port: COM6
=====
Serial Communication Example - Complete Version (Refactored)
=====
Port: COM6
Baudrate: 57600
Parameters: 8 data bits, No parity, 1 stop bit (8,N,1)
=====

Ready to open serial port COM6
-----
Use port COM6? [y/n] (default: y, press Enter to continue): |
```

The program will automatically:

- Detect all connected motor stations (excluding gateway)
- Assign station IDs based on the number of motors detected:
 - **1 motor**: Assigned to `STATION\_ID\_X`
 - **2 or more motors**: Smaller station ID → `STATION\_ID\_X`, Larger station ID → `STATION\_ID\_Y`
- Display station configuration information and wait for user confirmation
- Automatically skip dual motor control if only 1 motor is detected

```
Windows PowerShell
Opening serial port COM6...
✓ Serial port COM6 opened

Message Printing Configuration:
  Enable: Print all sent command messages and received response messages (for debugging)
  Disable: No message output (default, for normal use)
  [y] Enable [n] Disable (default)
Your choice (default: n):

✓ Serial connection established

=====
Auto-detected 2 motor(s): [5, 6]
=====
✓ Auto-assigned: STATION_ID_X = 5, STATION_ID_Y = 6

✓ Gateway online, found 2 device station(s) (excluding gateway)
  - Gateway (Station ID 3) is online
  - Device Station ID 5 is online ← Target
  - Device Station ID 6 is online

=====
Station Configuration Information:
  - Detected 2 motor(s)
  - STATION_ID_X = 5 (corresponds to station 5)
  - STATION_ID_Y = 6 (corresponds to station 6)
=====

Confirm station configuration and continue? [y/n] (default: y, press Enter to continue): |
```

Screenshot of the runtime error.

Check equipment power-up and CAN cable connection.

```

Windows PowerShell
Port: COM6
Baudrate: 57600
Parameters: 8 data bits, No parity, 1 stop bit (8,N,1)
=====
Ready to open serial port COM6
-----
Use port COM6? [y/n] (default: y, press Enter to continue):

Opening serial port COM6...
✓ Serial port COM6 opened

Message Printing Configuration:
  Enable: Print all sent command messages and received response messages (for d
  Disable: No message output (default, for normal use)
  [y] Enable [n] Disable (default)
Your choice (default: n):

✓ Serial connection established
✗ Timeout (1.0s) - No responses received

✗ SdkGetML failed - No response received
No response from device.

✗ NoResponseError: SdkGetML failed - No response received
Program terminated due to communication error.
✗ Serial connection lost: None

✓ Serial port COM6 closed
PS D:\Windows_Python_Serial_Messge_UIM2513_AN251219>

```

- Method 2: Using Python IDLE
 - Open Python IDLE (search for "IDLE" in the Start menu), open the Python file in IDLE, then press F5 to run.
- Method 3: Using an IDE (Integrated Development Environment)
 - You can use IDEs such as PyCharm, Visual Studio Code, or PyDev to write and run Python programs.
- Method 4: Double-click to run (if file association is configured)
 - If Python is properly installed and file association is configured, you can directly double-click a .py file to run it (but the window will close immediately after the program finishes running).

Additional Tips

- Node ID Management:
 - Always ensure that each motor on the network has a unique Node ID to avoid conflicts.
 - Test each motor individually after assigning its Node ID to confirm the configuration.
- Troubleshooting:
 - If the motors do not respond as expected, double-check the wiring, Node ID configuration, and uploaded code.
- Documentation:
 - Keep a record of each motor's Node ID for future reference.
 - Label the motors physically if managing multiple devices.

Remark

- Verify the supply voltage and current requirements for the UIM342 and stepper motors.

With the above steps, users can effectively control the UIM342 motor drive via UIM2513. The system is highly adaptable to a wide range of automation and robotic applications.